

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Big Data Analytics Digital Assignment -1

DA1: Topic Selection & Planning Report

Course Code: BCSE402L

Faculty: Dr. Dharini R. (E2+TE2)

Submitted By:

Pranav Satish Khadse - 23BLC1237
Rishi Jain - 23BLC1242
Anmol Kapil - 23BLC1229
Abhinav Kumar Singh – 23BLC1209

Predictive Shipment Delay Analytics for Logistics Decision Support

S.No	Name	Registration No.	Role
1	Pranav S. Khadse	23BLC1237	Team Coordinator / Data Engineering
2	Abhinav Kumar Singh	23BLC1209	Data Engineering (HDFS & Spark)
3	Rishi Jain	23BLC1242	Modeling & Hyperparameter Tuning
4	Anmol Kapil	23BLC1229	Explainability & Dashboard Development

1. Problem Statement

1.1 Domain and Dataset

The project is located in the logistics and supply chain domain, which is characterized by continuous, high velocity and high-volume transactional data generated throughout the procurement, warehousing and last mile delivery process. The team will use the *DataCo Smart Supply Chain for Big Data Analysis* dataset available in Kaggle, which has attributes that describe order dates, shipping modes, order delivery information, information about the product, information about the geography of customers, and the profitability of each order.

1.2 Problem Definition

The project aims to build a scalable Big Data Analytics pipeline that predicts shipment delays, explains the underlying causes of those delays using Explainable AI, and recommends optimal delivery buffer times to logistics providers.

1.3 Significance

Delayed shipments directly impact customer trust and lead to expensive processes of expedited shipping, penalty clauses and re-routing as a reaction. Most delay prediction attempts assume that the problem could be tackled by doing a small-scale machine learning on a static, pre-cleaned CSV file, which is not representative of the way that logistics data arrives in reality: continuously, in large volume and from multiple and diverse sources. The project also provides a clear and interpretable basis for logistics planners to modify delivery-time commitments, as predictions are not opaque but rather explicit: through a demonstration that delay prediction can be operationalised at a scale that a single-machine Pandas workflow is not able to maintain, the project illustrates that this is possible.

2. Course Relevance

The table below maps each Big Data Analytics concept covered in the course to its concrete application within this project.

Big Data Concept	Application in This Project
Hadoop HDFS	Distributed, fault-tolerant storage of the raw and intermediate shipment datasets, simulating how logistics providers persist high-volume order records.
Apache Spark (PySpark)	In-memory, distributed preprocessing and feature engineering (null handling, encoding, date-time feature extraction) across the full order dataset.
Spark SQL	Declarative, distributed querying for exploratory data analysis — aggregations by region, carrier, and product category prior to modeling.
NoSQL Concepts	Discussion of schema-flexible storage as an alternative for semi-structured shipment-tracking logs; used conceptually to justify design choices.
Machine Learning at Scale	Delay classification using Logistic Regression as a baseline, and XGBoost/LightGBM trained on Spark-processed feature sets.
Explainable AI (SHAP)	Post-hoc interpretation of model predictions to identify which features (carrier, distance, order season) drive delay risk.
Business Intelligence	Interactive dashboard (Streamlit / Power BI) translating model output into a decision-support tool for planners.

3. Explanation of Basic Concepts

The pipeline begins with data at rest in HDFS, which distributes file blocks across a cluster of nodes to provide fault tolerance and parallel read access. Apache Spark then operates on this data through Resilient Distributed Datasets and DataFrames, allowing preprocessing operations such as cleaning, joining, and feature construction to execute in parallel across partitions rather than sequentially on a single machine. Spark SQL is used for exploratory aggregation queries, since its Catalyst optimizer produces efficient execution plans over large tables without requiring the team to hand-tune query logic.

Once features are engineered, the modeling stage uses gradient-boosted tree ensembles (XGBoost and LightGBM) alongside a Logistic Regression baseline, with Optuna used for automated hyperparameter search. Because tree ensembles are inherently difficult to interpret, SHAP (SHapley Additive exPlanations) is applied to attribute each prediction to individual feature contributions, grounded in cooperative game theory. This closes the loop between raw distributed data and a decision a human planner can trust and act on.

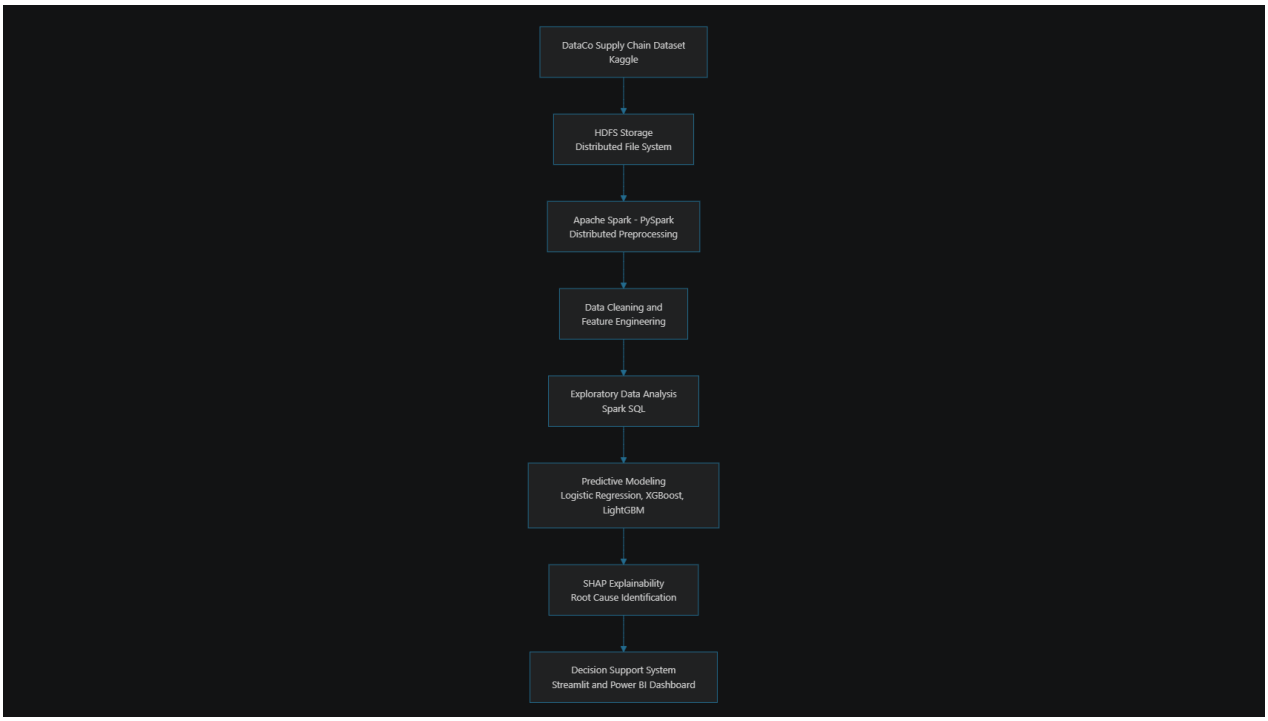


Fig 1. Big Data Processing Pipeline

The system architecture separates project into four layers — storage, distributed processing, analytics, and presentation — so that each component can be developed, tested, and scaled independently.

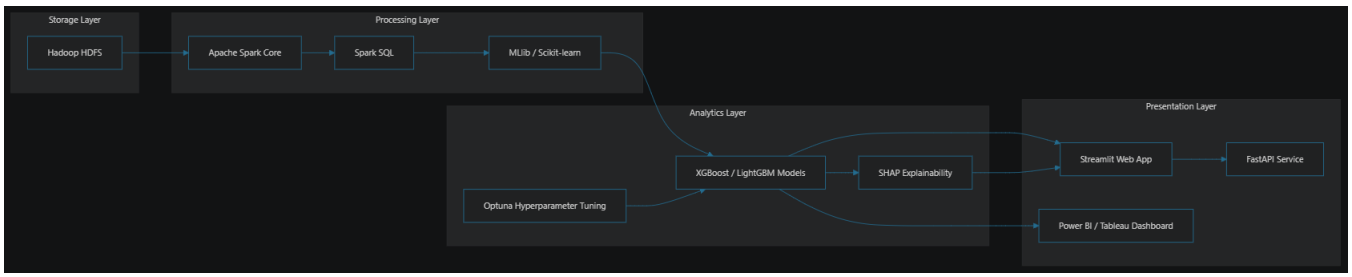


Fig 2. System Architecture

4. Outline of Approach & Roles

4.1 Methodology

The project follows a five-stage methodology: (1) data collection and ingestion into HDFS, (2) Spark-based preprocessing and feature engineering, (3) exploratory data analysis via Spark SQL and visualization libraries, (4) predictive modeling with explainability, and (5) dashboard-based presentation for decision support. Each stage produces a concrete artifact (cleaned dataset, EDA notebook, trained model, SHAP report, dashboard) that feeds into the next.

4.2 Timeline Across DA1–DA3

The schedule below aligns the project's technical milestones with the three graded deliverables of the course, so that DA2 reflects genuine mid-project progress and DA3 reflects a fully completed system rather than work compressed at the end.

Milestone	Activities	Target Window	Deliverable
DA1	Problem definition, dataset sourcing, environment setup (HDFS/Spark install), planning report	20 Jul – 4 Aug 2026	Planning Report
DA2	HDFS ingestion, Spark preprocessing, Spark SQL–based EDA, baseline model (Logistic Regression)	5 Aug – 19 Sep 2026	Progress Report
DA3	Advanced modeling (XGBoost/LightGBM), Optuna tuning, SHAP explainability, dashboard build, final report and slides	20 Sep – 20 Oct 2026	Final Report + PPT

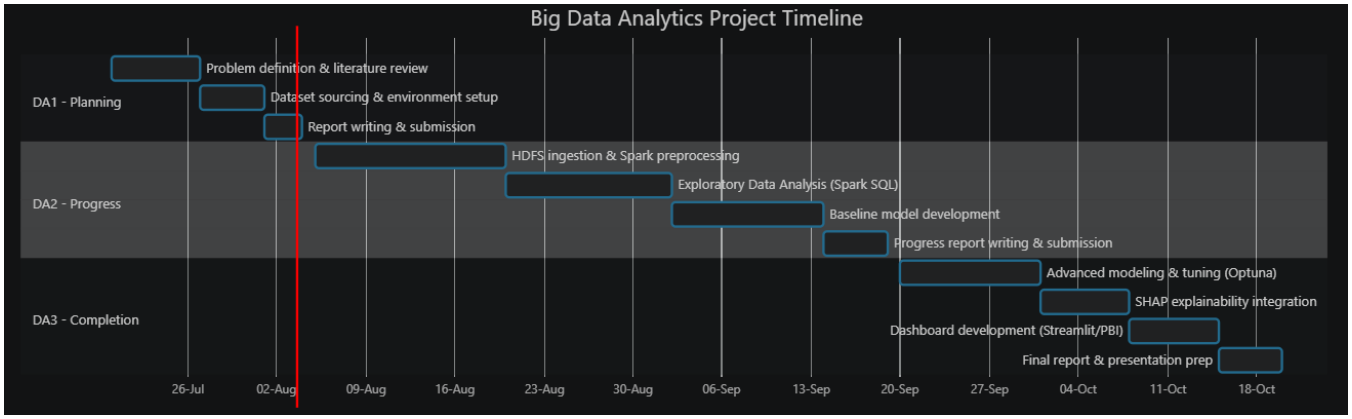


Fig 3. Gantt Timeline

4.3 Individual Responsibilities

Work is divided so that each member owns one stage of the pipeline end-to-end while remaining accountable to the team for integration points, keeping responsibility clear without creating hard hand-off dependencies.

Member	Primary Responsibility
Member 1 (Coordinator)	Overall project coordination, report consolidation, integration testing across pipeline stages, viva preparation.
Member 2	HDFS cluster setup, dataset ingestion, Spark preprocessing scripts, feature engineering.
Member 3	EDA via Spark SQL, model training (Logistic Regression, XGBoost, LightGBM), Optuna-based tuning.
Member 4	SHAP explainability analysis, Streamlit/Power BI dashboard, deployment via FastAPI.

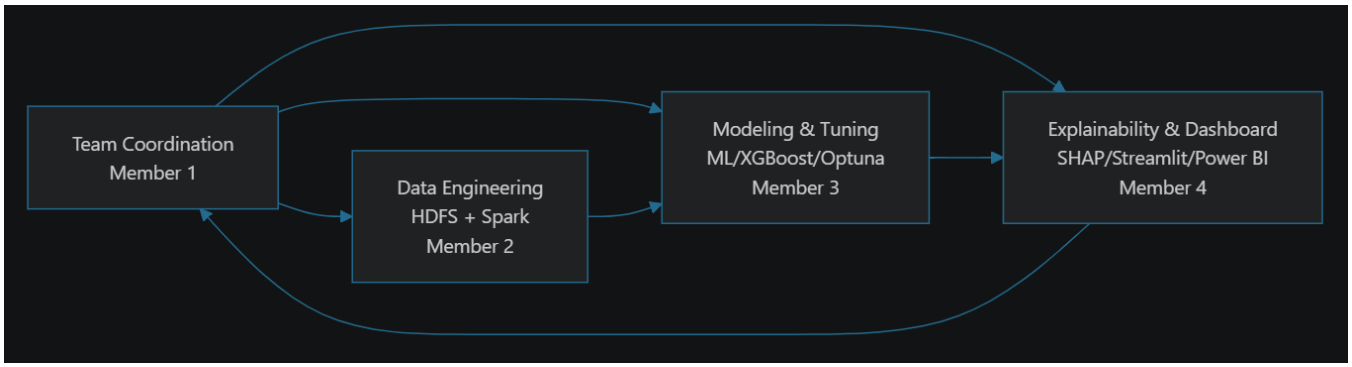


Fig 4. Team Role Distribution

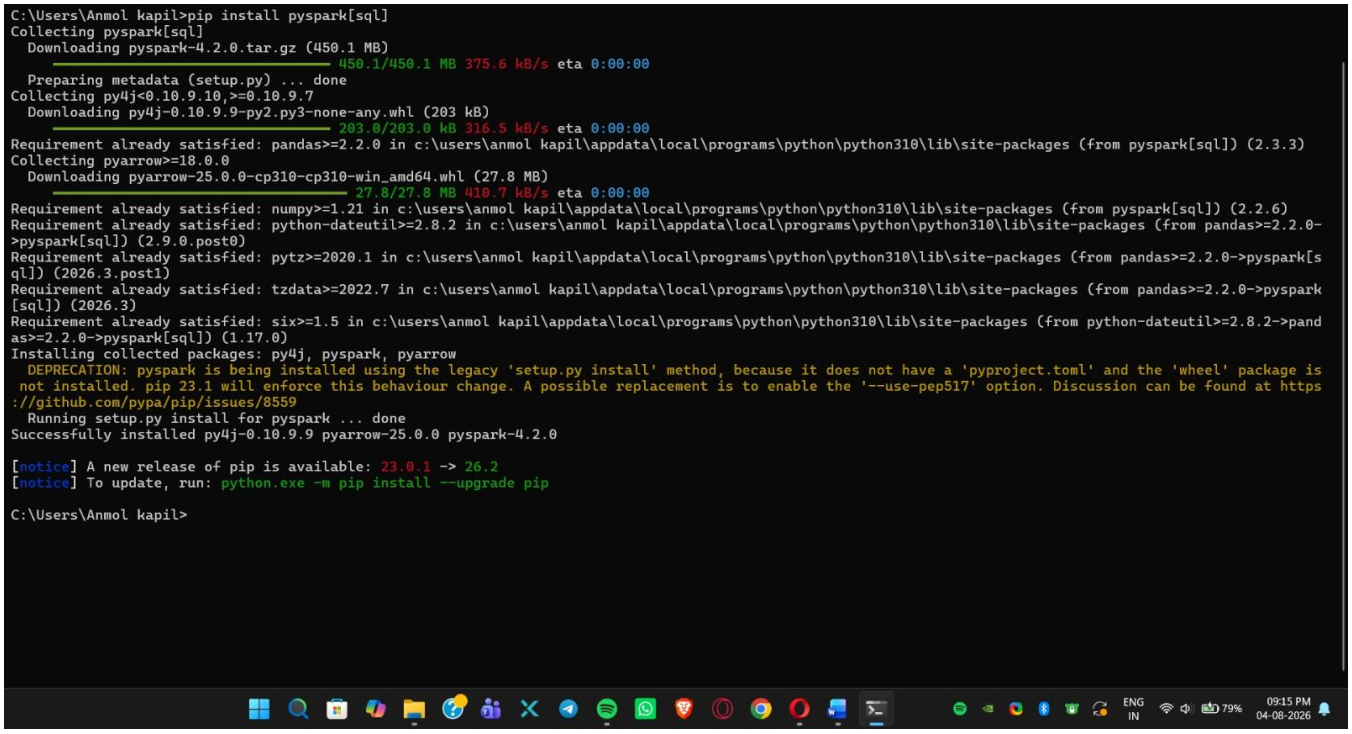
5. Demonstration of Feasibility

5.1 Dataset Source

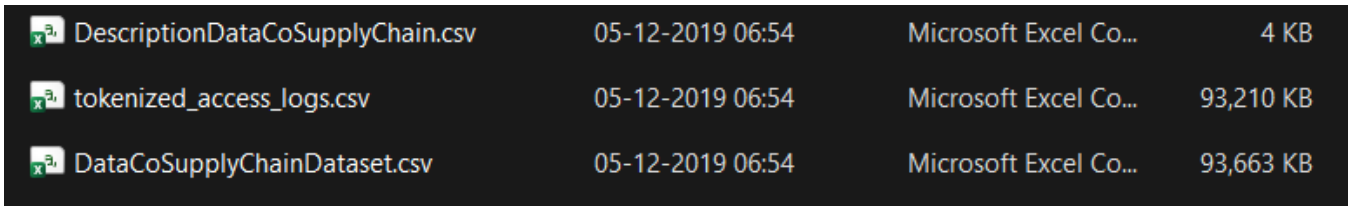
The DataCo Smart Supply Chain for Big Data Analysis dataset is publicly available on Kaggle and licensed for academic use. It has already been located and its schema reviewed, confirming the presence of the order-date, shipping-date, shipping-mode, and delivery-status fields required to construct the delay-prediction target variable.

5.2 Evidence of Setup

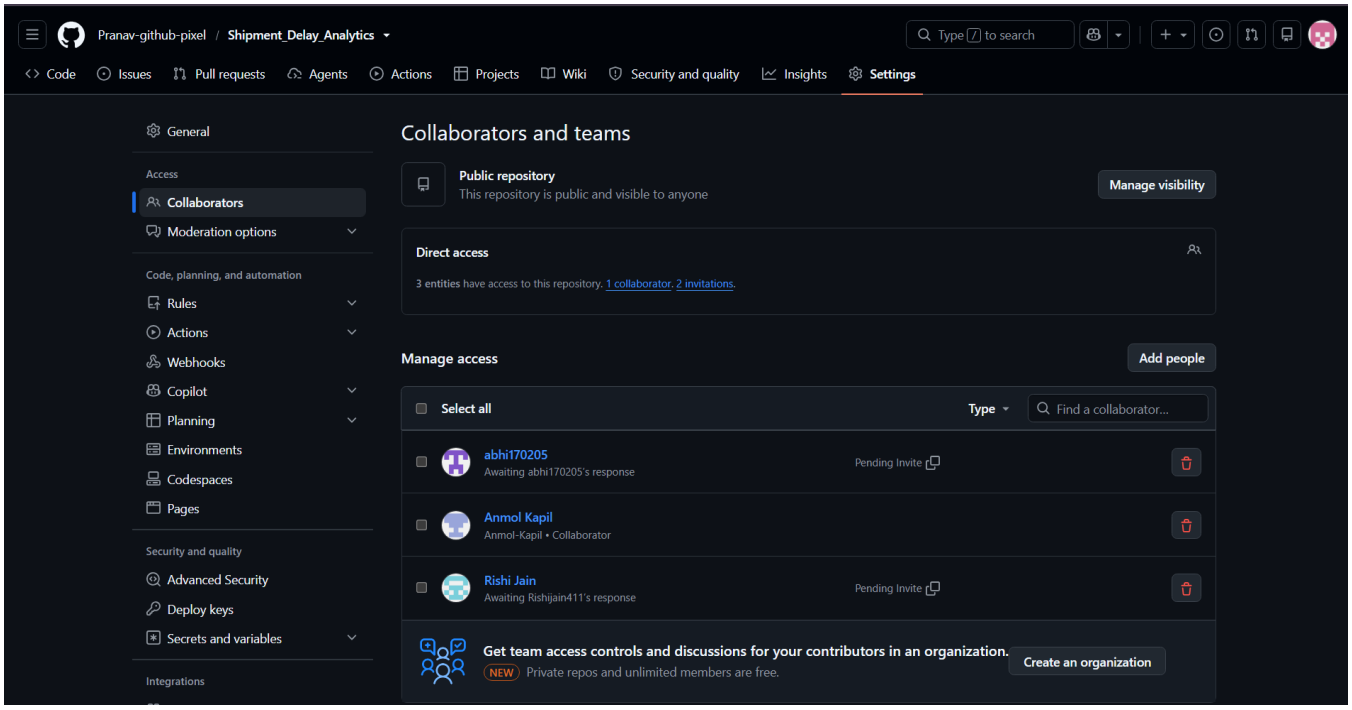
- Hadoop (pseudo-distributed mode) and Apache Spark installed and verified on the team's development environment.
- PySpark session tested with a sample read/write against a local HDFS instance.



- Dataset downloaded from Kaggle and a preliminary schema/row-count check completed (~180,000 records, 53 columns).



- Python environment configured with the core analytics stack: pandas, scikit-learn, xgboost, shap, optuna, streamlit.
- GitHub repository created for version control and team collaboration.



5.3 Expected Challenges and Mitigation

Expected Challenge	Mitigation Strategy
Local pseudo-distributed HDFS/Spark cluster has limited resources compared to production clusters	Use a representative data sample for iterative development; validate scalability logic conceptually and note cluster-scaling considerations in the report.
Class imbalance between delayed and on-time shipments	Apply class-weighting and resampling techniques (e.g., SMOTE) during model training, evaluated with precision/recall rather than accuracy alone.
Integrating SHAP with distributed-trained models	Use Spark for preprocessing and switch to a driver-local Pandas DataFrame for final model training and SHAP analysis, which is standard practice at this data scale.
Coordinating four-member contributions on a shared codebase	Maintain a shared GitHub repository with feature branches and weekly sync check-ins ahead of each DA submission.

6. Expected Outcome

- A trained model that predicts shipment delays with strong classification performance on held-out data.
- A ranked list of the factors most responsible for delay risk, derived from SHAP values.
- Carrier- and route-specific recommended delivery buffer times.
- An interactive dashboard that lets a logistics planner query predicted delay risk and root causes for a given shipment profile.
- A demonstrated, end-to-end integration of Big Data storage/processing, Explainable AI, and Business Intelligence.

7. Proposed Technology Stack

Component	Technology
Programming Language	Python
Big Data Storage	Hadoop HDFS
Distributed Processing	Apache Spark (PySpark), Spark SQL
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost, LightGBM
Hyperparameter Tuning	Optuna
Explainable AI	SHAP
Dashboard / BI	Streamlit, Power BI / Tableau
API Layer	FastAPI
Version Control	GitHub

8. References

[1] DataCo Smart Supply Chain for Big Data Analysis, Kaggle. https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

[2] Apache Software Foundation, "Apache Hadoop Documentation." https://hadoop.apache.org/docs/

[3] Apache Software Foundation, "Apache Spark Documentation." https://spark.apache.org/docs/latest/

[4] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," NeurIPS, 2017.